SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Student Name: S. Hari Prasath (192424423)

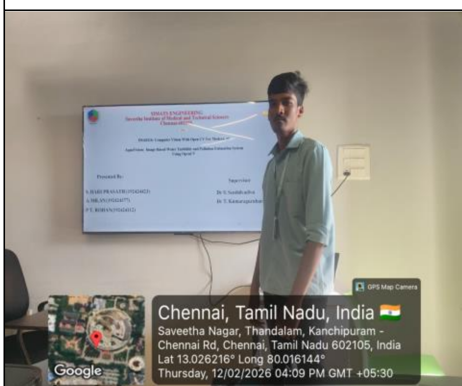
Course Code: DSA0216 **Slot: B**

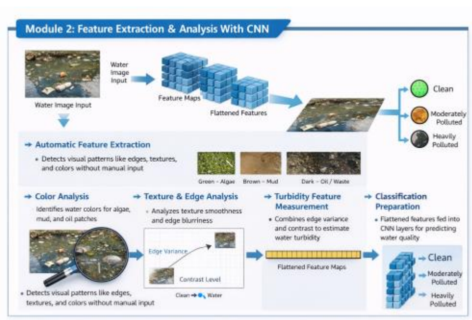
Course Name: Computer Vision with OpenCV for Modern AI

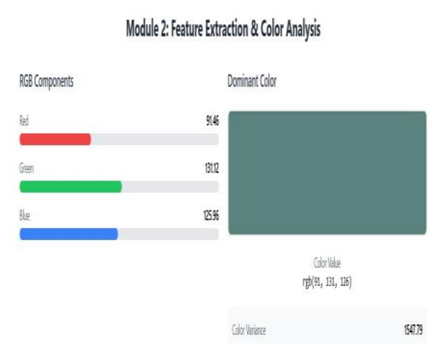
Course Faculty: Dr S. Senthilvadivu & Dr. T Kumaragurubaran

Project Title: Aqua Vision: Image-Based Water Turbidity and Pollution Estimation System Using OpenCV

Module Photographs:







Project Description: **Feature Extraction & Analysis**

Aqua Vision is a smart, image-based system that checks water quality by analysing photos of rivers, lakes, or seas using OpenCV and the Feature Extraction & Analysis module. You simply snap a picture with your phone, and it quickly tells if the water is clear or polluted without needing lab tests or costly sensors. Perfect for quick checks in places like Chennai's waterways. It works by first cleaning up the image removing blur, adjusting colours, and focusing on the water area. Then, the Feature Extraction module pulls out simple clues like muddy brown shades, cloudy textures, green algae hints, or floating trash patterns. These get turned into easy scores: turbidity from 0 to 100 , plus a pollution index. Results pop up as marked photos showing bad spots, charts of dirt levels, and alerts like "Unsafe for drinking." It uses basic math like colour histograms and edge detection no fancy AI needed at first. Runs on phones or web apps, saves 80% on monitoring costs, and helps track pollution trends over time. Great for environment lovers, local cleanup drives, or daily water safety in India.

Student Signature **Guide Signature**